Federation Hub Configuration Guide



Version 5.7

March 2026

Distribution Statement A: Approved for public release; distribution is unlimited.

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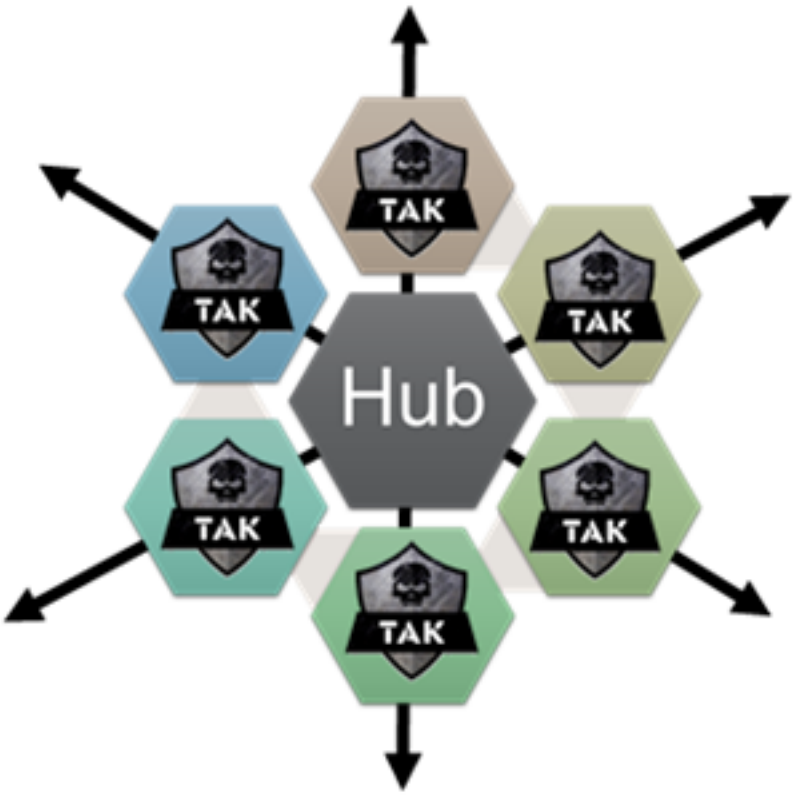
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1 About FedHub

The TAK Federation Hub (FedHub) enables forwarding of federated TAK data in a multi-hop setup. The FedHub supports centralized management of federation configuration using a hub and spoke topology. Instead of TAK Server federates connecting to each other directly, they connect to a FedHub. The FedHub performs connection and trust management, and brokers federated data according to a policy defined by an administrator. The FedHub software can be installed and run independently of TAK Server. It consists of three micro-services: a policy manager, an administrative UI, and a messaging broker. A system service manages starting and stopping these micro services.



1.0.1 Download

https://tak.gov/products/tak-server

2 Change Log

See https://wiki.tak.gov/display/DEV/TAK+Server+Federation+Hub+Change+Log

3 Installation

3.1 Prerequisites

3.1.1 System Requirements

3.1.1.0.1 Supported Operating Systems

- Red Hat Enterprise Linux (RHEL) 7/8

- Rocky 8
- CentOS 7/8
- Ubuntu 22/24

3.1.1.0.2 System Specs

- Java 17
- 4 processor cores
- 8 GB RAM
- 20 GB disk storage

3.1.2 Java Install / Upgrades

3.1.2.0.1 RHEL 7 or CentOS 7

```
sudo yum install -y https://download.oracle.com/java/17/latest/jdk-17_linux-x64_bin.rpm
```

```
java --version
```

This should tell you have 17.x.y. If the command tells you your Java version is not 17.x.y, then you can use the alternatives command to change it:

```
sudo alternatives --config java
```

3.1.2.0.2 Rocky 8, RHEL 8 or Centos 8

```
sudo dnf install java-17-openjdk-devel -y
```

3.2 RPM

3.2.1 RHEL 7 or CentOS 7

Update Package Manager

```
sudo yum update -y
```

Install Java 17

```
sudo yum install wget -y
```

```
sudo wget https://download.oracle.com/java/17/latest/jdk-17_linux-x64_bin.rpm
```

```
sudo yum install -y ./jdk-17_linux-x64_bin.rpm
```

3.2.1.0.1 Install RPM

```
sudo rpm -ivh takserver-fed-hub-*.noarch.rpm --nodeps
```

3.2.1.0.2 Install Mongo (Optional)

Mongo is only required when the Mission Federation Disruption Tolerance (MFDT) feature is enabled. MFDT will store and forward data sync feed (mission) changes in a FedHub that occur during periods of disconnectivity between a FedHub and other FedHubs, or TAK Servers connected to it.

Make sure /opt/tak/federation-hub/configs/federation-hub-broker.yml has your database credentials defined. Defaults will be generated otherwise.

Example credentials: dbUsername: martiuser dbPassword: pass4marti

Install Commands:

```
sudo cp /opt/tak/federation-hub/scripts/db/mongodb-org.repo
    /etc/yum.repos.d/mongodb-org.repo

sudo yum install -y mongodb-org

sudo systemctl daemon-reload

sudo systemctl enable mongod

sudo systemctl restart mongod

sudo /opt/tak/federation-hub/scripts/db/configure.sh
```

3.2.1.0.3 Start Fedhub

```
sudo systemctl daemon-reload

sudo systemctl restart federation-hub

sudo systemctl enable federation-hub
```

3.2.2 Rocky Linux 8 or RHEL 8

3.2.2.0.1 Update Package Manager

```
sudo dnf update -y
```

Install Java 17

```
sudo dnf install java-17-openjdk-devel -y
```

Install RPM

```
sudo dnf install takserver-fed-hub-*.noarch.rpm -y
```

3.2.2.0.2 Add and Apply SELinux

```
sudo dnf install checkpolicy

cd /opt/tak/federation-hub && sudo ./apply-selinux.sh && sudo semodule -l | grep
    takserver
```

3.2.2.0.3 Install Mongo (Optional)

Mongo is only required when the Mission Federation Disruption Tolerance (MFDT) feature is enabled. MFDT will store and forward data sync feed (mission) changes in a FedHub that occur during periods of disconnectivity between a FedHub and other FedHubs, or TAK Servers connected to it.

Make sure /opt/tak/federation-hub/configs/federation-hub-broker.yml has your database credentials defined. Defaults will be generated otherwise.

Example credentials: dbUsername: martiuser dbPassword: pass4marti

Install Commands:

```
sudo cp /opt/tak/federation-hub/scripts/db/mongodb-org.repo
    /etc/yum.repos.d/mongodb-org.repo

sudo yum install -y mongodb-org

sudo systemctl daemon-reload

sudo systemctl enable mongod

sudo systemctl restart mongod

sudo /opt/tak/federation-hub/scripts/db/configure.sh
```

3.2.2.0.4 Start Fedhub

```
sudo systemctl daemon-reload

sudo systemctl restart federation-hub

sudo systemctl enable federation-hub
```

3.3 DEB

3.3.1 Ubuntu

(supported for version 5.1 and later) #### Update Package Manager

```
sudo apt update -y
```

Install Deb

```
sudo apt install <absolute path>/takserver-fed-hub_*.deb -y
```

3.3.1.0.1 Install Mongo (Optional)

Mongo is only required when the Mission Federation Disruption Tolerance (MFDT) feature is enabled. MFDT will store and forward data sync feed (mission) changes in a FedHub that occur during periods of disconnectivity between a FedHub and other FedHubs, or TAK Servers connected to it.

Make sure /opt/tak/federation-hub/configs/federation-hub-broker.yml has your database credentials defined. Defaults will be generated otherwise.

Example credentials: dbUsername: martiuser dbPassword: pass4marti

3.3.1.0.2 Install Commands:

```
sudo apt install curl software-properties-common gnupg apt-transport-https
    ca-certificates -y

curl -fsSL https://pgp.mongodb.com/server-7.0.asc |  sudo gpg -o
    /usr/share/keyrings/mongodb-server-7.0.gpg --
```

dearmor

```
echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ]
    https://repo.mongodb.org/apt
```

/ubuntu jammy/mongodb-org/7.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list

```
sudo apt update && sudo apt install mongodb-org -y
```

```
sudo systemctl enable mongod
```

```
sudo systemctl restart mongod
```

```
sudo /opt/tak/federation-hub/scripts/db/configure.sh
```

3.3.1.0.3 Start Fedhub

```
sudo systemctl daemon-reload

sudo systemctl restart federation-hub

sudo systemctl enable federation-hub
```

3.4 Docker

3.4.1 Prerequisites

- All commands should be run from the top level *takserver-fedhub-docker* folder
- Before using the Fedhub, you must setup the certificates for secure operation. If you have existing certs, you can copy them into the `tak/federation-hub/certs/files/` directory located within the *takserver-fedhub-docker* release folder. Otherwise, there will be steps later to generate them once the container is running.
- Make sure `tak/federation-hub/configs/federation-hub-broker.yml` has the database credentials defined that you would like to use. (DB is Optional)
  - Example:
    - `dbUsername: martiuser`
    - `dbPassword: pass4marti`

3.4.2 Network

This Docker network will be used to enable communication between the fedhub and databases containers. To create the network run:

```
docker network create fedhub-"$(cat tak/version.txt)"
```

3.4.3 Database

Make sure `tak/federation-hub/configs/federation-hub-broker.yml` has your database credentials defined!

3.4.3.0.1 Build Database Image

```
docker build -t fedhub-db:"$(cat tak/version.txt)" -f docker/Dockerfile.fedhub-db .
```

3.4.3.0.2 Run Database Image

The Fedhub database container can be configured to persist data directly to the host or only within the container.

- To persist to the host, create an empty host directory (unless you have a directory from a previous docker install you want to reuse). For upgrading purposes, we recommend that you keep the database directory outside of the docker release folder. Replace `<path>` with the absolute path to db host directory

```
docker run -d -v <path>:/var/lib/mongodb:z -v
    $(pwd)/tak/federation-hub/configs:/opt/tak/federation-hub/configs:z -it -p
    27017:27017 --network fedhub-"$(cat tak/version.txt)" --network-alias hub-db --name
    fedhub-db-"$(cat tak/version.txt)" fedhub-db:"$(cat tak/version.txt)"
```

- To run Fedhub database with container only persistence

```
docker run -d -v $(pwd)/tak/federation-hub/configs:/opt/tak/federation-hub/configs:z -it
    -p 27017:27017 --network fedhub-"$(cat tak/version.txt)" --network-alias hub-db
    --name fedhub-db-"$(cat tak/version.txt)" fedhub-db:"$(cat tak/version.txt)"
```

3.4.4 Fedhub

3.4.4.0.1 Build Fedhub Image

```
docker build -t fedhub:"$(cat tak/version.txt)" -f docker/Dockerfile.fedhub .
```

3.4.4.0.2 Run Fedhub Image

```
docker run -d -v $(pwd)/tak:/opt/tak:z -it -p 9100:9100 -p 9101:9101 -p 9102:9102
    --network fedhub-"$(cat tak/version.txt)" --name fedhub-"$(cat tak/version.txt)"
    fedhub:"$(cat tak/version.txt)"
```

3.4.5 Certificates

The Federation Hub authenticates connections using TLS with X.509 certificates.

You may copy existing certs into place (`tak/federation-hub/certs/files` for example) or generate new ones.

Ensure that the certificate configuration paths in `tak/federation-hub/configs/federation-hub-broker.yml` and `tak/federation-hub/configs/federation-hub-ui.yml` match the location where you generated / copied your certs to.

If you did not copy over existing certs as noted above, you will need to generate some now:

1. Edit `tak/certs/cert-metadata.sh`
2. Generate root ca

```
docker exec -it fedhub-"$(cat tak/version.txt)" bash -c "cd /opt/tak/certs &&
    ./makeRootCa.sh"
```

3. Generate server cert

```
docker exec -it fedhub-"$(cat tak/version.txt)" bash -c "cd /opt/tak/certs &&
    ./makeCert.sh server takserver"
```

4. Generate client cert

```
docker exec -it fedhub-"$(cat tak/version.txt)" bash -c "cd /opt/tak/certs &&
    ./makeCert.sh client <username>"
```

5. Set Authorized User

```
docker exec -it fedhub-"$(cat tak/version.txt)" bash -c "java -jar
    /opt/tak/federation-hub/jars/federation-hub-manager.jar
    /opt/tak/certs/files/<my_client_cert.pem>"
```

6. Restart Fedhub to load new certificates (make sure tak/federation-hub/configs/federation-hub-broker.yml matches your cert names and locations)

```
docker restart fedhub-"$(cat tak/version.txt)"
```

Full Cerfiticate Information Can Be Found Here: Certificate Generation

4 Configuration

4.1 Ports

The Federation Hub by default uses TCP ports:

- 9100 - Admin Interface (UI)
- 9101 - V1 Federation (Deprecated)
- 9102 - V2 Federation

These ports can be changed if needed in:

`/opt/tak/federation-hub/configs/federation-hub-broker.yml`

`/opt/tak/federation-hub/configs/federation-hub-ui.yml`

4.2 Firewall

You may need to configure your firewall to open the ports that the Federation Hub uses. On CentOS, you can add the required firewall rules with the following commands (assuming defaults):

```
sudo firewall-cmd --zone=public --add-port 9100/tcp --permanent

sudo firewall-cmd --zone=public --add-port 9102/tcp --permanent

sudo firewall-cmd --reload
```

Note for cloud users: you will need to configure security groups and ingress/egress rules for the ports as well.

4.3 Certificates

The Federation Hub authenticates connections using TLS with X.509 certificates. Scripts for generating a private security enclave, including a Certificate Authority (CA), and certs for use by the Federation Hub are in the TAK Server documentation here: Certificate Generation.

The instructions for doing so are the same as for TAK Server **except** all of the certificate generation scripts are located in and should be run from `/opt/tak/federation-hub/certs` instead of `/opt/tak/certs`. You may also copy existing certs into place rather than generating new ones.

Ensure that the certificate configuration paths in `/opt/tak/federation-hub/configs/federation-hub-broker.yml` and `/opt/tak/federation-hub/configs/federation-hub-ui.yml` match the location where you generated / copied your certs to.

4.3.0.1 Restart Fedhub

```
sudo systemctl daemon-reload

sudo systemctl restart federation-hub
```

4.4 Authorized Users

The Federation Hub authenticates clients using TLS with X.509 client certificates. To authorize clients to act as administrators and enable access to the admin UI, use federation-hub-manager.jar:

```
sudo su tak
java -jar /opt/tak/federation-hub/jars/federation-hub-manager.jar
    /opt/tak/federation-hub/certs/files/<admin>.pem
exit
```

By default, authorized users are written to `/opt/tak/federation-hub/authorized_users.yml`. Optionally, you can specify the location of the output file where the authorized users are written to:

```
sudo su tak
java -jar /opt/tak/federation-hub/jars/federation-hub-manager.jar
    /opt/tak/federation-hub/certs/files/<admin>.pem ./authorized_users.yml
exit
```

The location of the authorized users file should match the location configured for the admin UI microservice (in `/opt/tak/federation-hub/configs/federation-hub-ui.yml`).

4.5 Files

The Federation Hub has two main configuration files: - The Broker microservice configuration is used for configuring the server as it relates to federation and messaging - `/opt/tak/federation-hub/configs/federation-hub-broker.yml`

- The UI microservice configuration is used for configuring access to the UI
  - `/opt/tak/federation-hub/configs/federation-hub-ui.yml`

NOTE: The default Federation Hub configuration files assume a *takserver.jks* server keystore and a *fed-truststore.jks* truststore, as these are the defaults when setting up TAK Server from the configuration guide. If you are using different keystore or truststore names, both files should be edited to change the keystore and truststore files and passwords to be correct for your setup. Also ensure that the keyAlias in the federation-hub-ui.yml file matches the `<name>` you used when generating the server cert (`./makeCert.sh server <name>`).

4.6 Keycloak

Federation Hub supports Oauth admin authentication using keycloak. This can be configured by adding the following properties to the `/opt/tak/federation-hub/configs/federation-hub-ui.yml` file.

```
allowOauth: true
oauthPort: 9099 # make sure this gets updated / matches the port in keycloakrRedirectUri
keycloakServerName:
keycloakDerLocation: /opt/tak/federation-hub/certs/keycloak-certs.der
keycloakClientId:
keycloakSecret:
keycloakrRedirectUri: https://localhost:9099/login/redirect # make sure this matches the
    oauthPort
keycloakAuthEndpoint:
keycloakTokenEndpoint:
keycloakAccessTokenName:
keycloakRefreshTokenName:
keycloakClaimName: groups        # the keycloak token attribute to look at for
    authenticating
keycloakAdminClaimValue: admin   # the keycloakClaimName value to match for verifying
    admin status
```

4.7 Tokens

TAK Server / Federation Hub have the ability to authenticate federation connections via a token rather than x509 mutual authentication. While tokens are going to be less secure than traditional x509 mutual auth (mTLS), the benifit of them is that they will allow federation connections to work in network environments where mutual authentication is not an option. (such as networks running break and inspect on all tls connections)

To open a port that supports token authentication, configure your `/opt/tak/federation-hub/configs/federation-hub-brok` file: (port can be any open port, type should be `jwt`)

```
federationTokenAuthServers:
  - type: jwt
    port: 9103
```

There are two different types of token authentication that federate clients can use to connect: - **Automatic** is enabled on a per CA basis. When you have a CA Group node on the graph, edit the node and enable the *Allow Token Auth* option. This will allow incoming federate connections to present thier client certificates over TLS (instead of traditional mTLS) to authenticate and automatically receive a token. - **Manual** is enabled by dragging a Token Group node onto the graph and generating tokens. These tokens are then manually distributed to other TAK admins so they can drop them into their outgoing connection configuration for connecting to the Fedhub. (you technically don't need the remote side's ca.pem when using manual tokens, but they still need yours)

5 How to Use UI

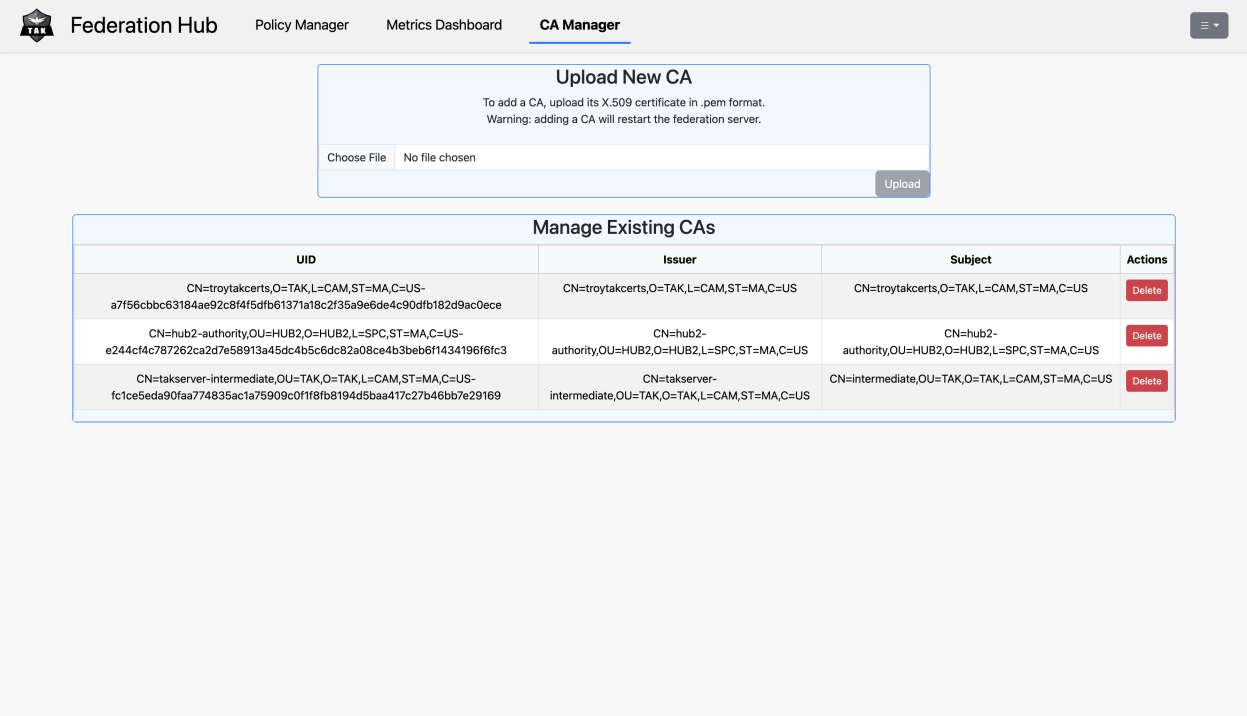
5.1 Certificate Manager

The Federationn Hub performs connection and trust management by primarily using x509 mutual authentication (this is the most secure option, however we also support token authenticaiton as well).

In order for the federation hub to trust other federate servers and vice versa, they must share each others certificate authorities (CAs) in order to create a federate trust-store. See Settings to download the federation hub ca.pem. It can also be found at `/opt/tak/federation-hub/certs/files/ca.pem`. TAK Server admins can find their ca.pem at `/opt/tak/certs/files/ca.pem`

To import the ca.pem of a federate server that is going to connect to the hub, select the "Choose File" button, navigate to and select the .pem file, and press "Upload." The CA will be added to the truststore.

Once a CA has been imported it will be available as an option on the policy graph.

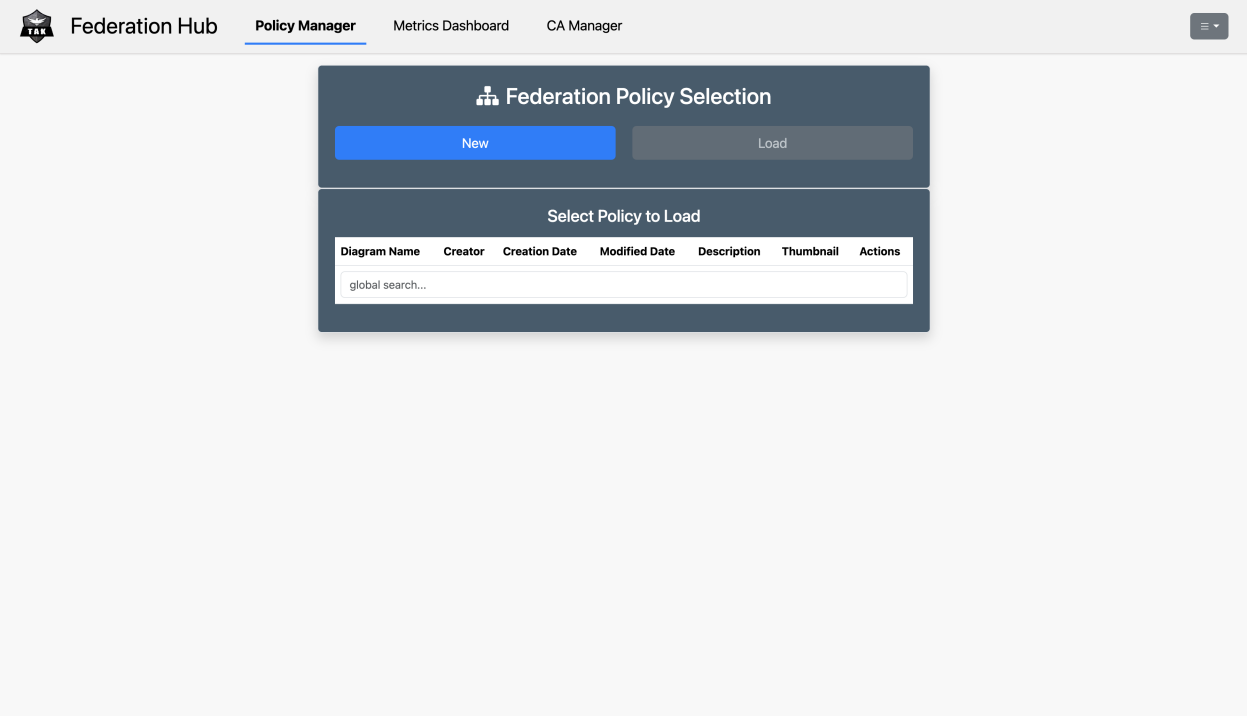


5.2 Policy Manager

In a Web browser, navigate to https://localhost:9100/index.html (or the IP address/domain your Federation Hub is reachable at). To authenticate to the Federation Hub, use the same certificate that was authorized in the Configuring Authorized Users steps.

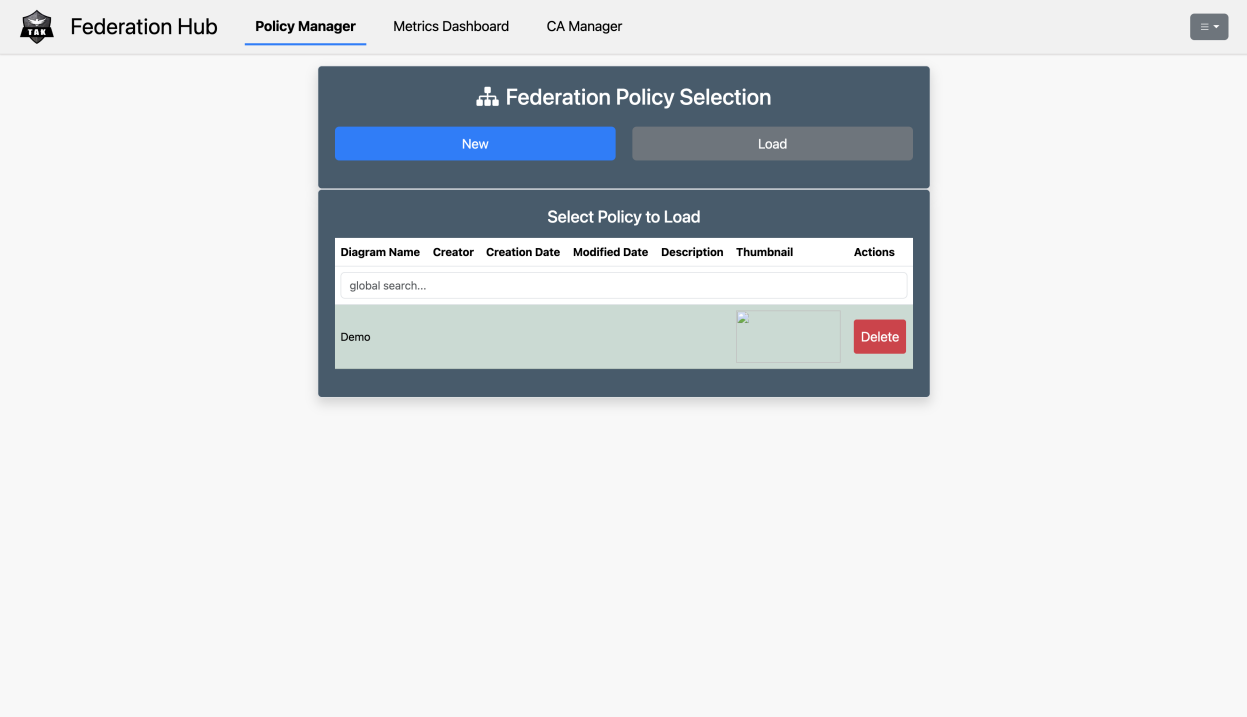
5.2.1 Creating a Policy

Once authenticated, you will see this page:



Select the “New” option under Federation Policy and fill in the form to create a new blank policy.

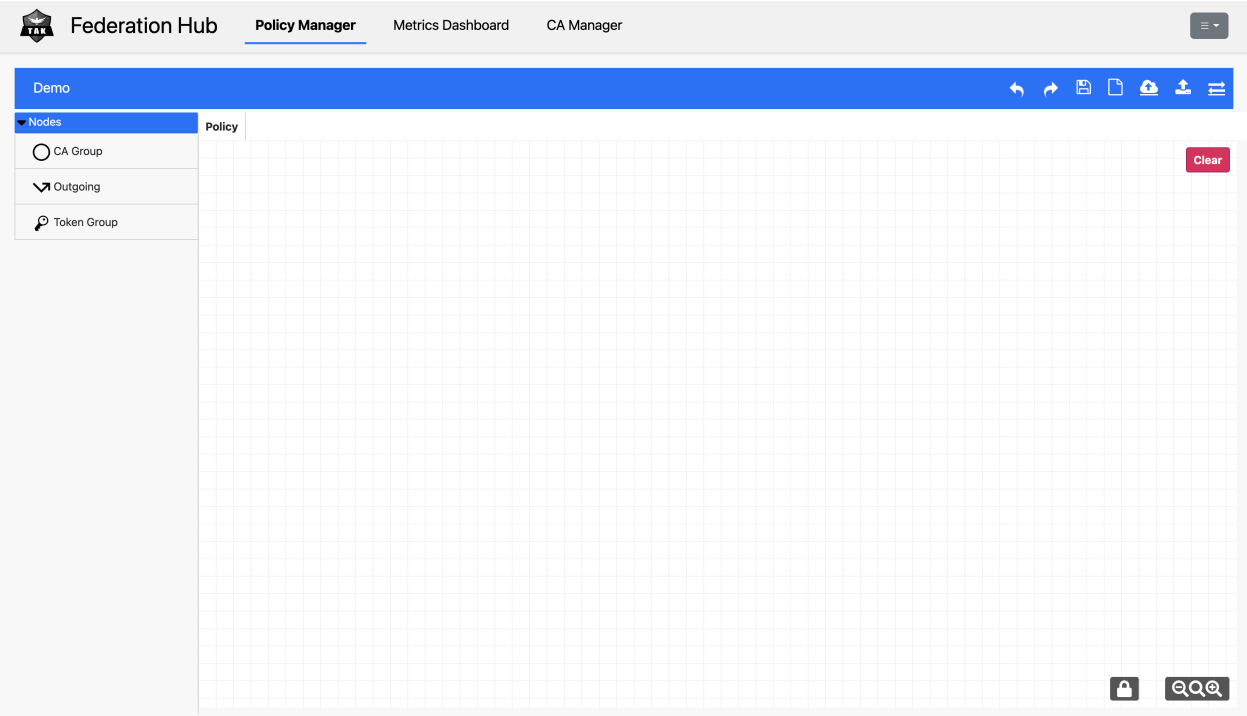
5.2.2 Loading a Policy



Existing policies will be displayed in the table and can be loaded with the “Load” option after selecting a policy in the table.

5.3 Policy Editor

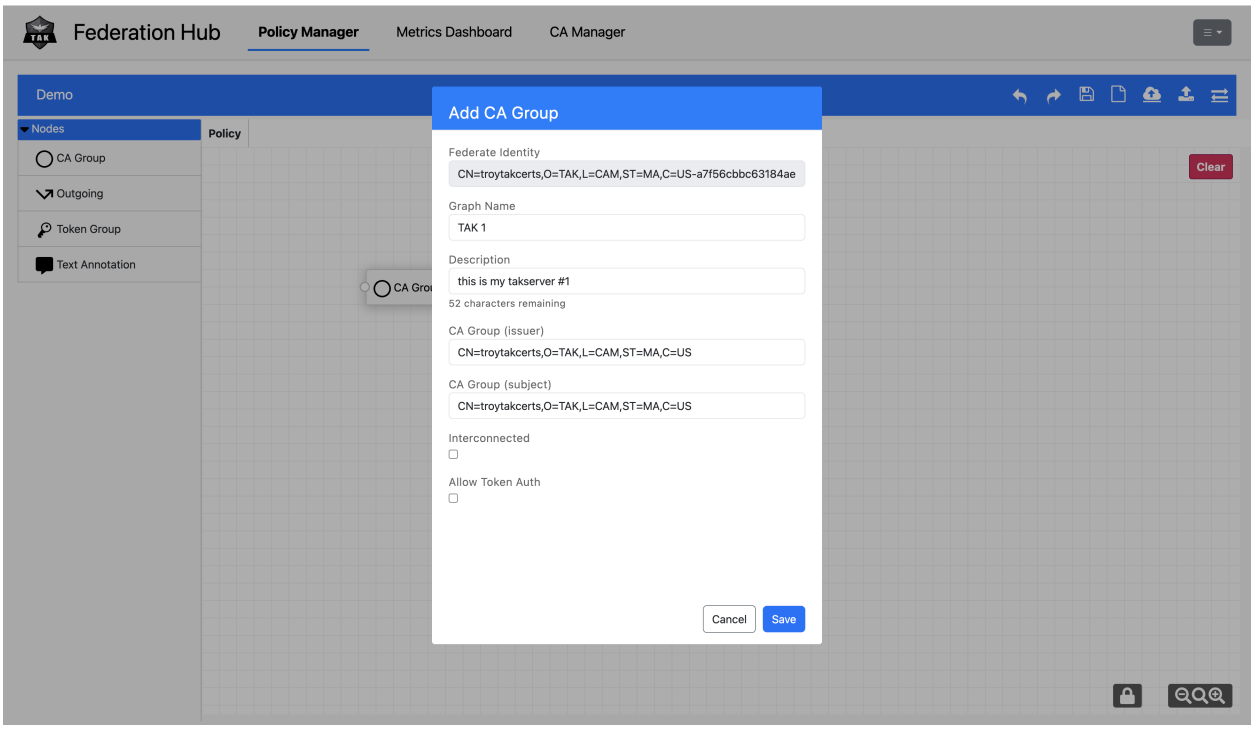
Once you have created a new policy or loaded an existing one, you will be brought to the graph editor. This is where you will manage how the federates connecting to the Fedhub are allowed to interact with each other.



5.3.1 Add / Modify CA Group Node

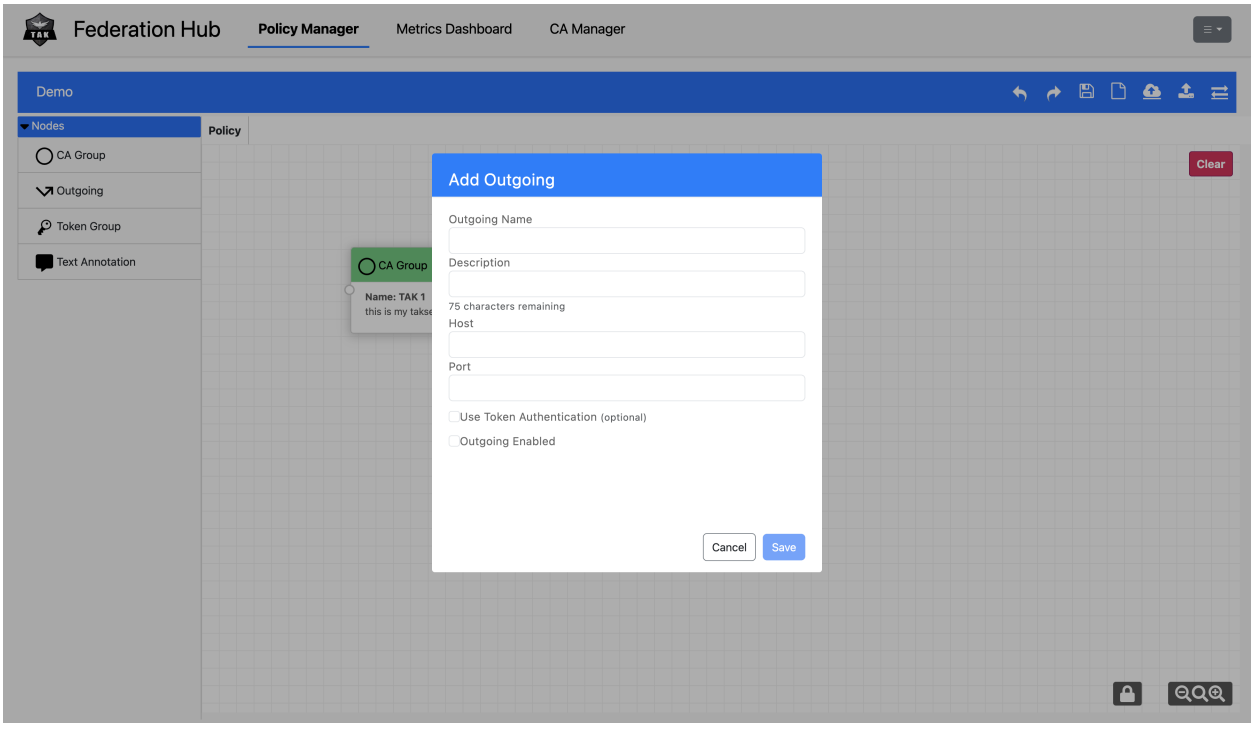
A Certificate Authority (CA) group represents the CA of a TAK Server/Fedhub that wants to connect to this Fedhub. To add a CA Group to the graph simply drag a "CA Group" Node onto the canvas. You will be presented with a form. You will want to browse the list of available CA's by either their issuer or subject name and select the one you want to configure.

You will see two checkboxes at the bottom. - Enabling **"Interconnected"** means that if multiple servers are connected to the hub and they are using the same CA, they will be automatically linked together and therefore share all of their data with each other. (Enabling this is very uncommon as it is recomended to run one set of unique certs per server) - **Allow Token Auth** When enabled, will automatically allow incoming federate connections to present thier client certificates over TLS (instead of traditional mTLS) to authenticate and receive a token. This will allow federation connections to work in network environments where mutual authentication is not an option (Ex. if your network is running break and inspect on all tls connections). The Token Auth Duration is how often the clients token will expire forcing them to reconnection. A value of -1 means the token will not expire. Any value greater than 0 may cause disruptions in service until the remote end attemps to reconnect.



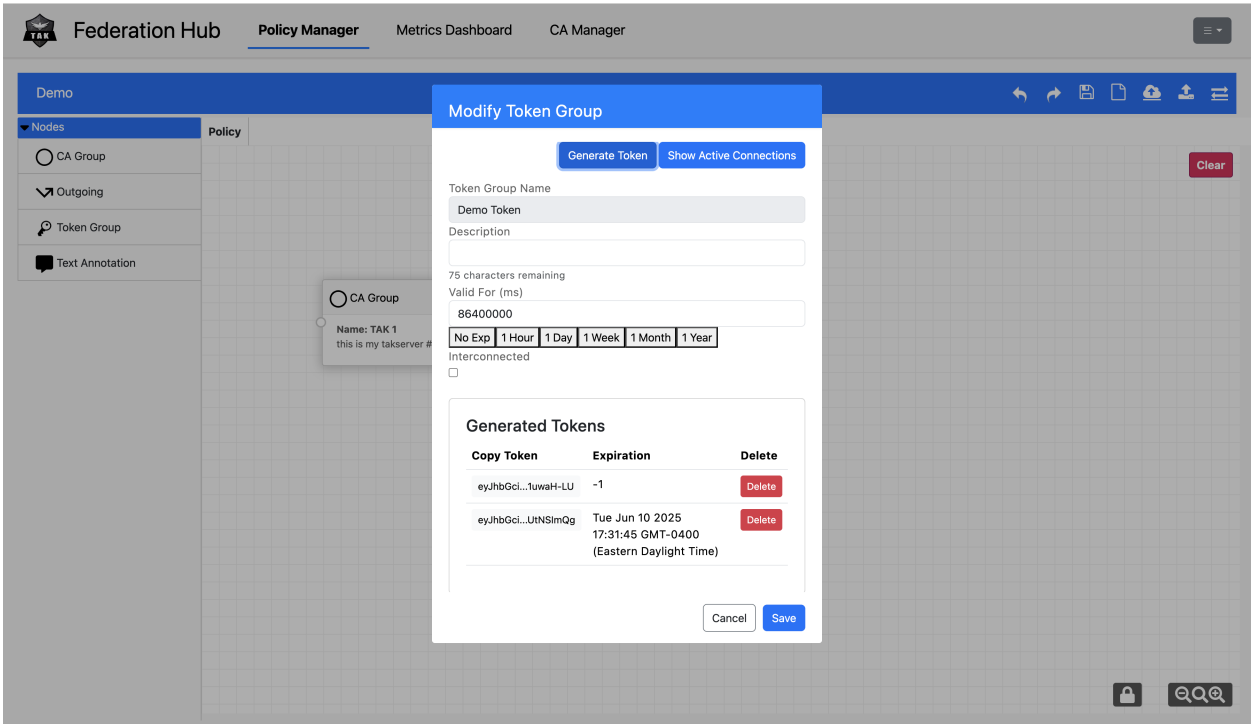
5.3.2 Add / Modify Outgoing Node

An Outgoing Connection node will establish a connection from this Fedhub to another TAK Server or Fedhub. To add an Outgoing Connection to the graph simply drag a "Outgoing" Node onto the canvas. You will be presented with a form. Enter the connection information of the other server and select the enabled checkbox when you are ready for the connection to be initiated.



5.3.3 Add / Modify Token Group Node

A Token group represents a set of token(s) that TAK Servers/Fedhubs can use to connect to this Fedhub. To add a Token Group to the graph simply drag a "Token Group" Node onto the canvas. You will be presented with a form. Here you can generate and manage your active tokens. A value of -1 means the token will not expire. The difference between a token group and the CA Group token auth mentioned above is that the CA Group is automatically given a token by presenting it's cert. In this case for a Token Group, it is more of a manual process as you will need to send the other server's admin a new token when theirs expires.



5.3.4 Connections Groups

Next we will connect these CA groups / token groups / outgoing connections to each other with "edges" to begin federating data. Edges are unidirectional so to do this, drag from the left or right of the source node and complete the drag on the destination node. The edge arrow will indicate the flow of traffic. When the edge between nodes is created, a form will appear and ask you how you want to configure group filtering.

5.3.4.1 Filtering options:

- **Allow All Messages** This must be used for all V1 server connections and any V2 server connections that aren't running group mapping. This will allow all messages without performing any filtering
- **Allowed Groups** Allow any message that contains at least one of these groups
- **Disallowed Groups** Block any message containing one of these groups
- **Allowed And Disallowed Groups** Allow messages that contains an allowed group and contains no disallowed groups

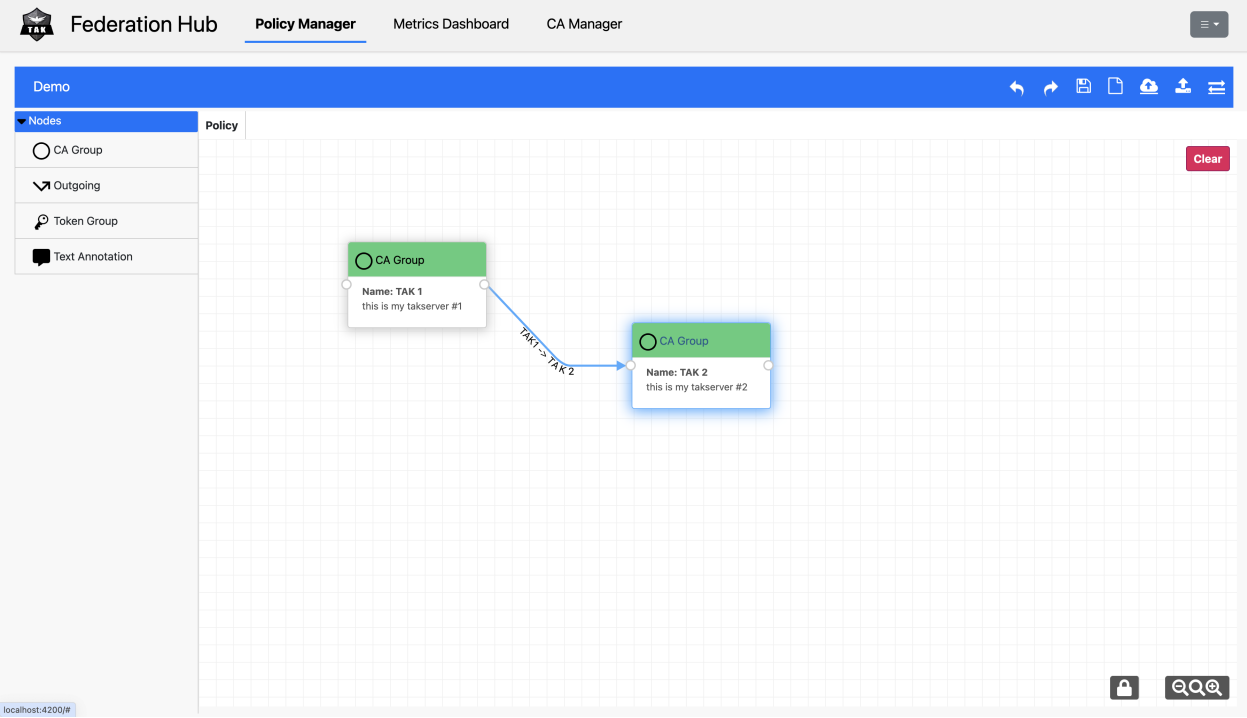
Note: The source node *must* be running group mapping to use a filtering option other than *Allow All Messages*

5.3.5 Update a Policy

Each time you add a node to the graph, delete a node, edit a node, add a connection, generate tokens or make any other changes on the graph you will need to save and upload the policy for it to 1. Save your changes and 2. Take effect. To save the policy and set the updates as active, click the cloud icon on the

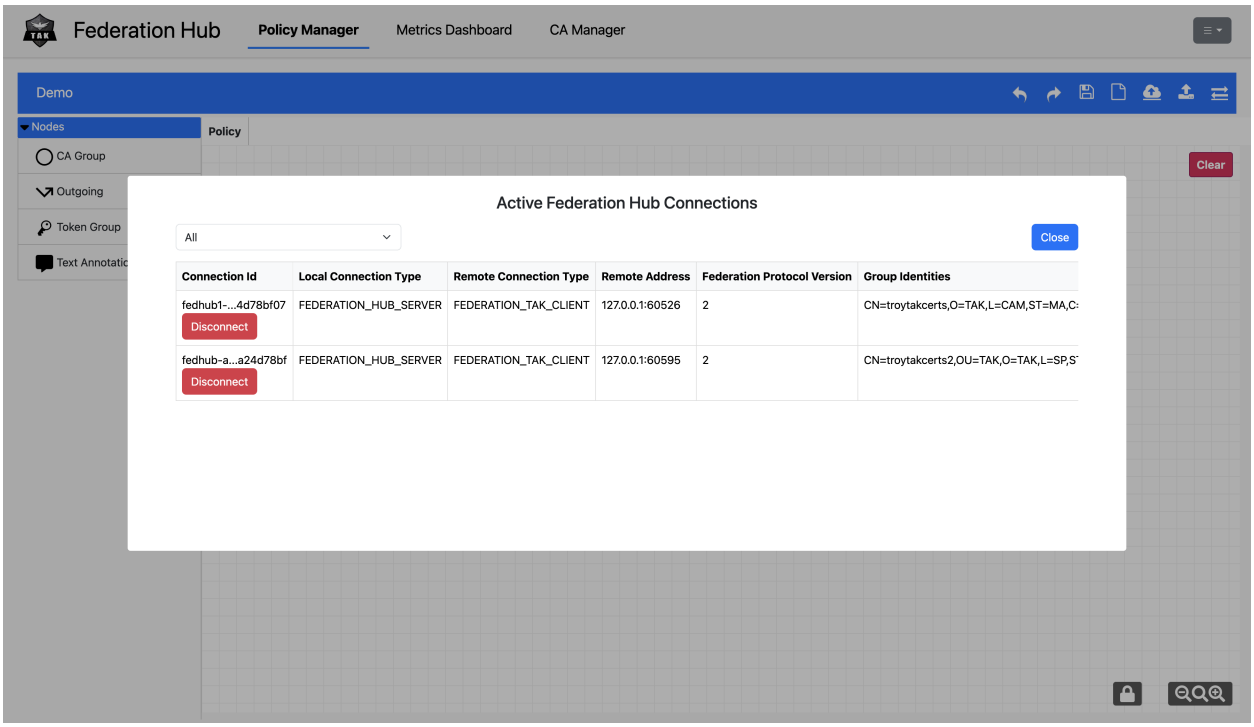
top toolbar. Once the policy is active, federates can connect to the Hub. The Federation Hub will deny connections from servers that are not present in the Hub's policy.

A Successful connection to the Hub will be indicated by a green node as seen below



5.4 Active Connections

A List of active connections can be viewed by clicking the double arrow icon on the toolbar or by double clicking a node and selecting the "Show Active Connections" option



5.5 Metrics Dashboard

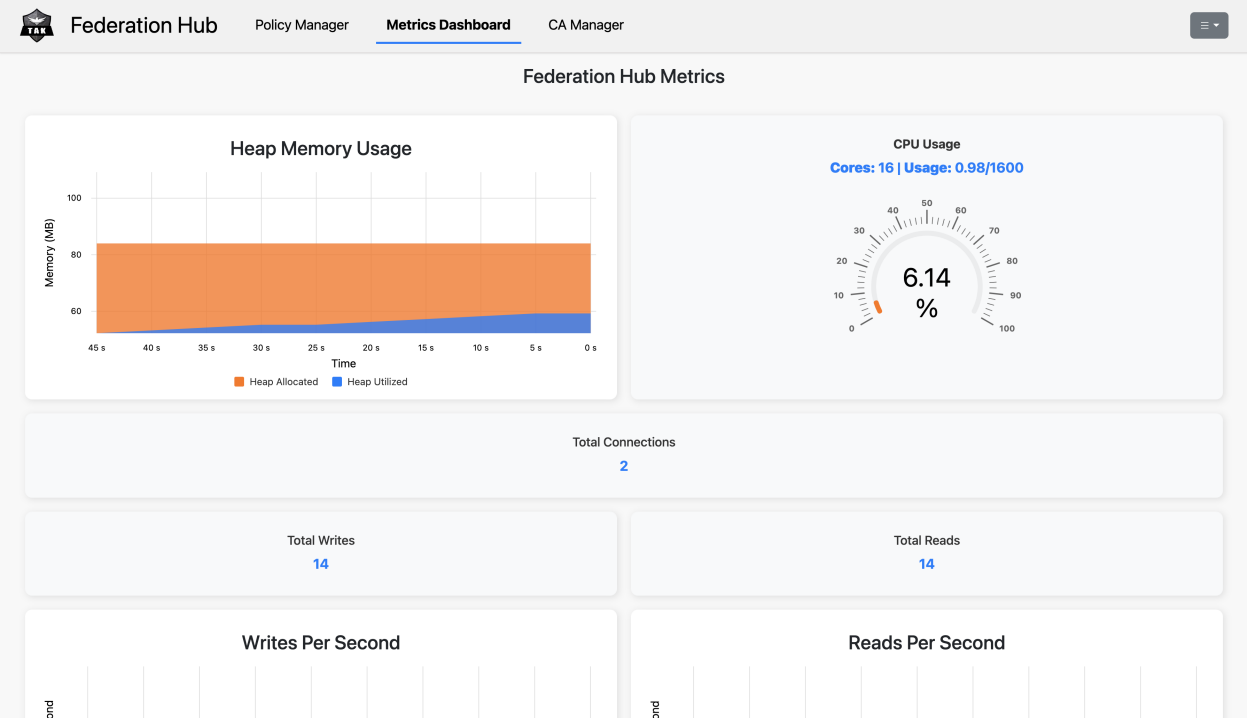
The Fedhub Metrics Dashboard is available in the Metics tab. The dashboard continuously renders the following information:

Heap Usage Fedhub runs inside one or more Java Virtual Machines (JVM). Heap Commited is how much heap memory in MB is allocated and Heap Used is how much of that is currently being used.

CPU Usage How much of the CPU of the machine the server is running on is currently being used.

Total Connections This tells you how many connections are established to / from Fedhub

Network I/O and Reads/Writes This tells you how much TCP and UDP traffic the server is currently handling, as well as a brief history.



5.6 Settings

The settings page currently offers two options - **Download Selft CA** - this will allow you to download your ca.pem file from the browser rather than through the command line interface - **Restart Broker** - this will restart the Fedhub's broker service

